higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

15 April 2025

This marking guideline consists of 7 pages.

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NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
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SECTION A

QUESTION 1: FLOW MEASUREMENT

1.1 A rectangular weir has a notch of simple rectangular shape.✓ A V-notch weir has a triangular notch, customarily measuring either 60 or 90 degrees.✓ Cippoletti weir is much like a rectangular weir, except that the vertical sides of the notch have slope of approximately a 14 degree angle from vertical.✓ (3)

1.2
- Flumes
- Weirs (2)

1.3
1.3.1 D
1.3.2 C
1.3.3 A
1.3.4 C
(4 × 2) (8)

1.4 An orifice tab is welded on the orifice plate which extends out of the line✓ giving an indication of the orifice plate.✓ (2)

1.5 Bernoulli theory has illustrated that the flow through a restriction is not directly to differential pressure across the restriction.✓ but varies directly as the square root of different pressure.✓ The flow can be determined by taking the square root of the pen.✓ (3)
[18]

QUESTION 2

m = 0,32
C =0,608✓ from the graph

$$E = \frac{1}{\sqrt{1-0,32^2}}$$✓

$$= 1,0555$$✓

$$m = 0,32$$
SHR =1,4

$$\frac{h}{P_a} = \frac{\Delta P}{P_{¿|+P_{atm}¿}} \text{ X } 27,2$$

$$\frac{h}{P_a} = \frac{1}{3,6764+101,3} \text{ X } 27,2$$✓

$$= 0,26$$✓

$$\varepsilon = 0,969$$✓ from the graph

$$R_d = \frac{3,54\,Q\rho}{\mu D\sqrt{m}}$$

μ(viscosity)= $1,31\,x\,10^{-4}$ at 30℃

$$R_d = \frac{3.54 \times 740 \times 6,408}{1,31\,x\,10^{-4} \times 250 \times \sqrt{0,32}}$$✓

$$= 90\ 609,936$$✓

$Z = Z_R \text{ x } Z_D$
Z_R = 1,001 ✓with m = 0,32 and Rd = ± 90 000
Z_D = 1,001✓ with D =250 mm and m = 0,32
$Z = Z_R \text{ x } Z_D$
= 1,001 x 1,001
=1,002✓

$$d = ¿\sqrt{\frac{740}{0,01252 \times 0,608 \times 1,002 \times 0,969 \times 1,0555\sqrt{\frac{102}{6,408}}}}$$✓✓

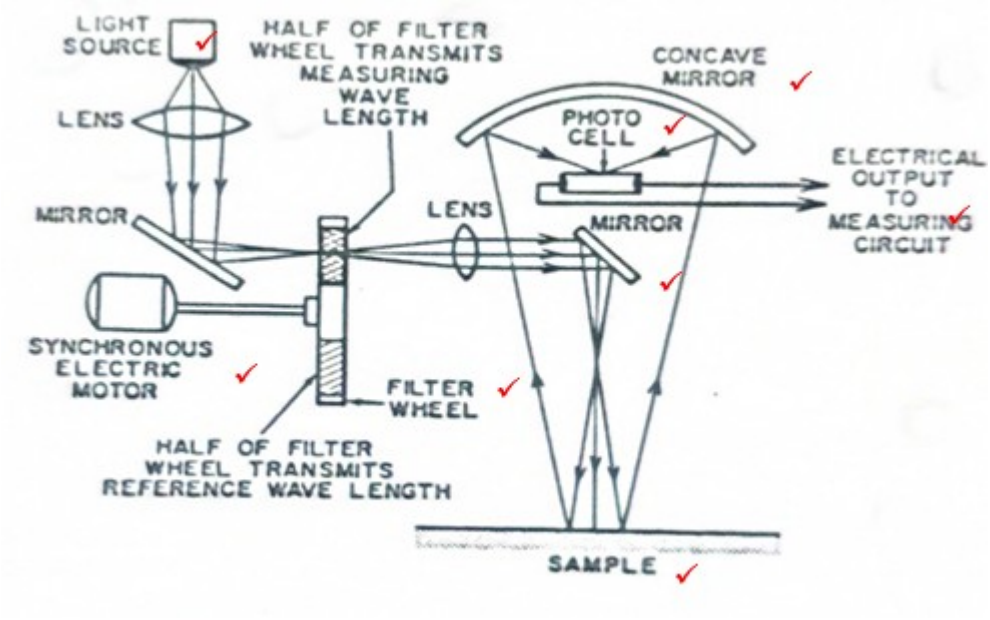
= 154,194 mm✓ **[12]**

TOTAL SECTION A: 30

SECTION B

QUESTION 3: DENSITY, HUMIDITY &VISCOSITY MEASUREMENT

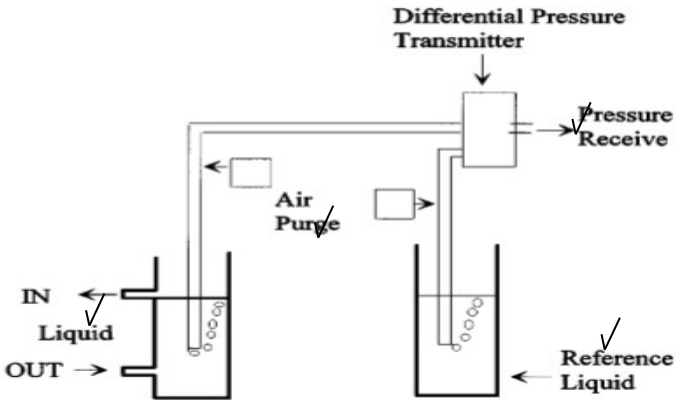
3.1



(7)

3.2
- The reference filter will measure a specific wavelength and will be compared with a measured wavelength obtained from the measuring lens.
- These wavelengths will be picked up by the photocell and induce two current pulses.
- The relation between the two signals will be a function of the vapour present on the surface of the sample. (3)

3.3 3.3.1

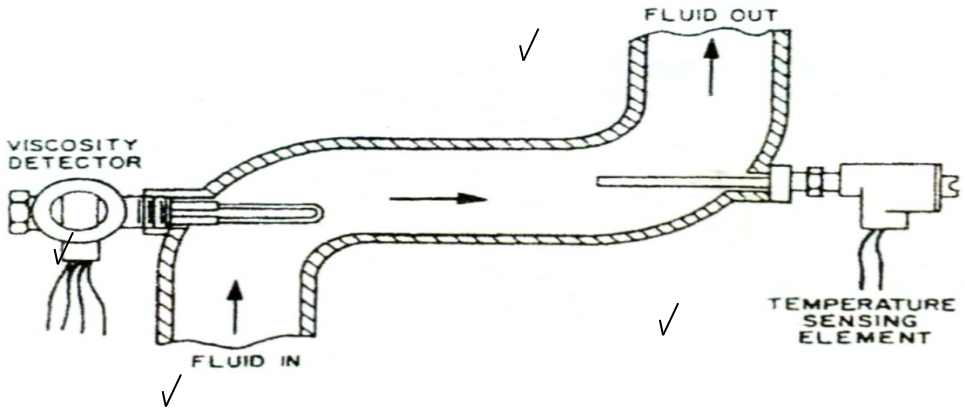


(4)

3.3.2 Both tubes must be placed at the same head, because P_1 must be equal to P_2 when the densities are the same.✓ When an unknown density must be measured the pressure at the same head will differ from the reference density.✓ If P_1 is higher than P_2 the bubble will appear at different times because more air pressure is needed at P_1 to overcome the density pressure.✓ The differential pressure will be directly proportional to the unknown density.✓ (4)

3.4 Viscosity refers to a fluid's resistance to flow.✓ Factors such as temperature,✓ pressure or shear rate,✓ and composition or size and shape of particles that make up the fluid.✓ (3)

3.5



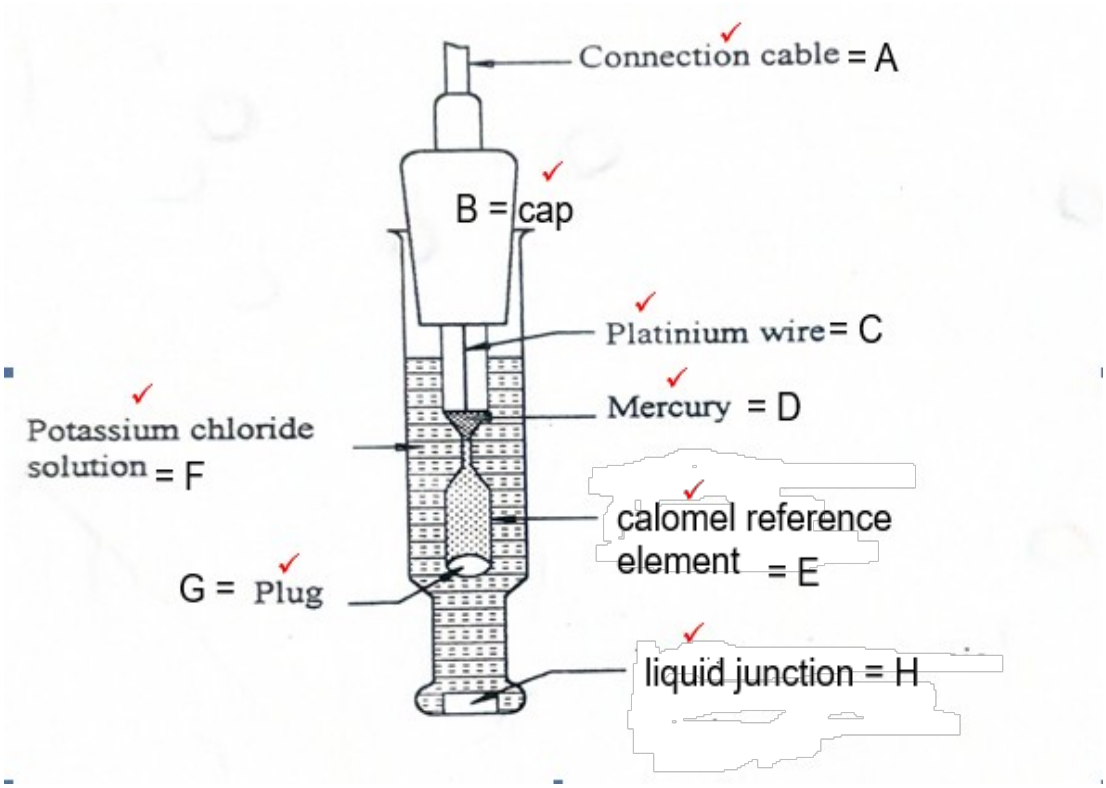
(4)
[25]

QUESTION 4: PH AND CONDUCTIVITY MEASUREMENT

4.1 pH of a solution is a measure of the acidity or alkalinity of a solution.✓
pH meters measure the potential difference between the reference electrode and the measuring electrode.✓✓ (3)

4.2 4.2.1 True
4.2.2 True
(2 × 1) (2)

4.3



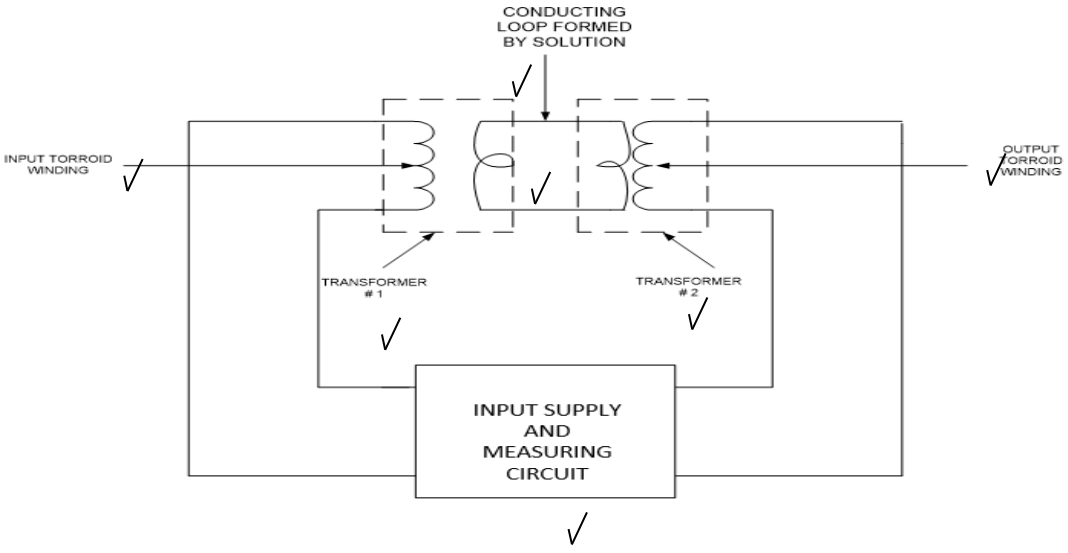
(8)

4.4
- When an electrolyte is decomposed by an electric current,✓ the mass of substance set free at an electrode✓ is proportional to the quantity of the electricity which flows through the solution.✓✓

OR

- When the same quantity of electricity is passed through solutions of different electrolytes✓, the mass of an ion discharge at an electrode✓ is proportional to its relative molecular mass divided by the number of charges on the ion.✓✓ (4)

4.5

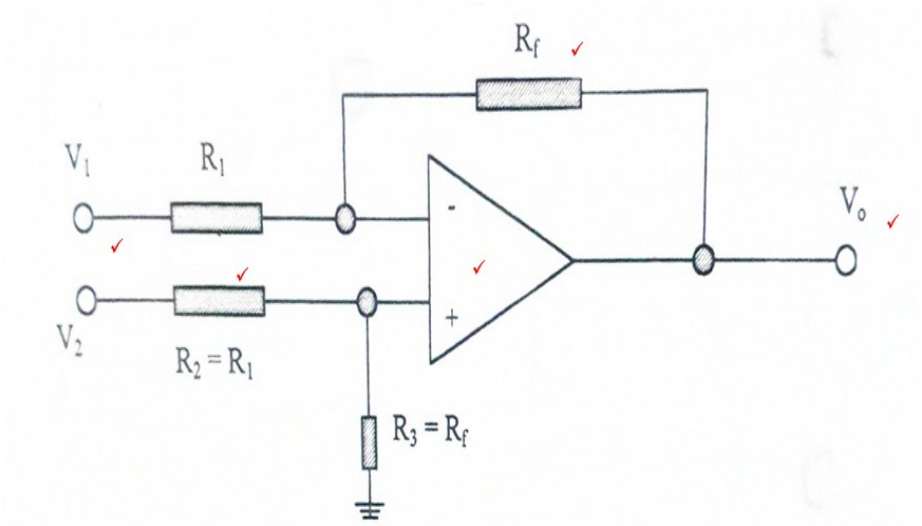


(7)
[24]

QUESTION 5: AUTOMATIC CONTROL

5.1 A controlled variable is the monitored or measured output variable✓ from a process that must be controlled within set limits.✓ Manipulated variable is the input variable.✓ (3)

5.2 5.2.1



(5)

5.2.2 The circuit is used to monitor the difference between two voltage levels.✓ It compares an input voltage with a reference voltage.✓ The output of the comparator indicates whether the input signal is above or below the reference voltage.✓✓ (4)

5.2.3 Wheatstone bridge✓
Linear variable differential transformer✓
Linear variable capacitive transducer✓ (3)

5.3 5.3.1 The time required for the integral action to repeat the change caused by the proportional action.✓✓

5.3.2 The time required for the proportional action to maintain the value and size of the derivative action✓ when the rate adjust valve is fully closed.✓

(2 × 2) (4)

5.4 Integral action is also referred to as reset action✓ while derivative action is known as rate action.✓ (2)

[21]

TOTAL SECTION B: 70
GRAND TOTAL: 100

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